

TrES-3b

R-1906

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140514 · created 2026-08-05T23:15:52+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456791.7515 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.01
Transit depth (Rp/Rs)^2	3.76 +/- 0.4 [%]
Orbital Inclination (inc)	82.73 +/- 0.49
Airmass coefficient 1 (a1)	1.222 +/- 0.019
Airmass coefficient 2 (a2)	-0.1593 +/- 0.0059
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	None
Optimal Aperture	3.67
Optimal Annulus	12.51
Transit Duration (day)	0.0648 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.194 ± 0.01 vs 0.16309 (deviation 2.81σ)
Duration fit vs published	0.0648 d vs 0.0567 d (deviation 2.11σ)
Mid-time O–C	-2.5 min
Depth SNR	17.7
Residual scatter	1.53 %

Light curve

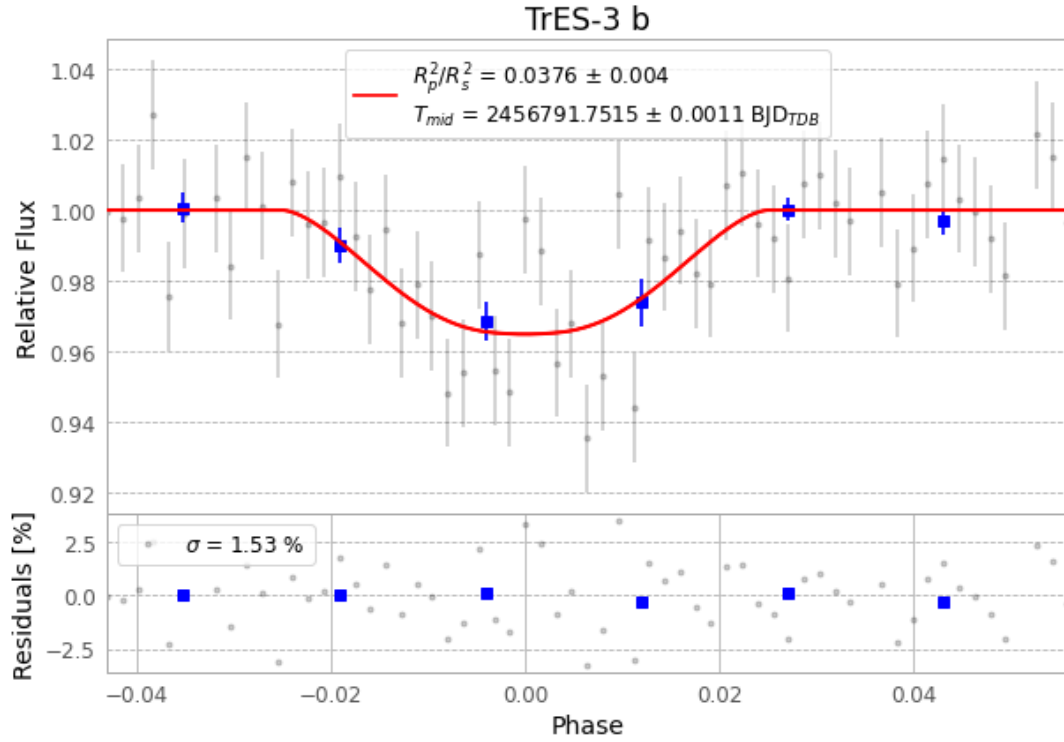


Figure 1 — FinalLightCurve_TrES-3 b_2014-05-13.png (SHA-256 2774f02c504d3adf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [192, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 84501eda19a0ad18...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), TRES-3140514043612.FITS → TRES-3140514074212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140514.log (21cb14bf81ea...), FinalLightCurve_TrES-3 b_2014-05-13.png (2774f02c504d...), FinalLightCurve_TrES-3 b_2014-05-13.csv (e15f2d2cf651...)

Compute resources

Wall time 146.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-05 23:15Z.